Exercises for the Cervical Region

Isometric Exercises—Self-Resistance

The intensity of the isometric exercises can be from light to strong, depending on the patient's symptoms and tolerance.

Patient position and procedure: Sitting.

- ***Flexion.*** Have the patient place both hands on the forehead and press the forehead into the palms in a nodding fashion while not moving (Fig. 16.37A).

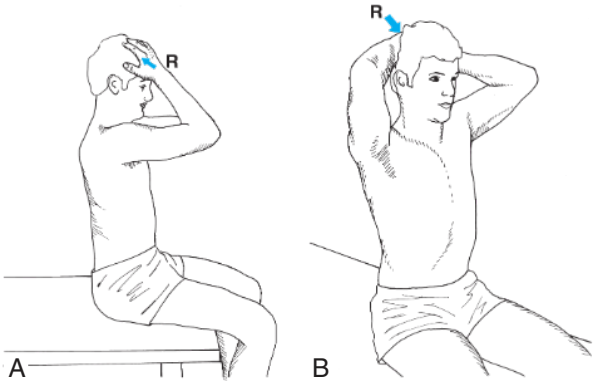


FIGURE 16.37 Self-resistance for isometric (*A*) cervical flexion and (*B*) axial extension.

- ***Side bending.*** Have the patient press one hand against the side of the head and attempt to side-bend, as if trying to bring the ear toward the shoulder but not allowing motion.
- ***Axial extension.*** Have the patient press the back of the head into both hands, which are placed in the back, near the top of the head
- ***Rotation.*** Have the patient press one hand against the region just superior and lateral to the eye and attempt to turn the head to look over the shoulder but not allowing motion.

Isometric Resistance Activities

- *Patient position and procedure:* Standing with a basketball-sized inflatable ball between the forehead and a wall. Have the patient keep the chin tucked and not go into a forward-head posture. The patient maintains the functional position while superimposing arm motions. Progress by adding weights to the arm motions.
- *Patient position and procedure:* Supine with the head over the edge of the mat, the neck maintained in a neutral functional position, and no support to the head. The patient must be able to keep the neck in its safe, functional position to perform this advanced stabilization exercise. He or she holds the position as tolerated. Progress by adding arm motions, then adding weights to the arm motions as tolerated.

Dynamic Cervical Flexion

Often with faulty forward-head postures, the patient substitutes using the sternocleidomastoid (SCM) muscles to lift the head when getting up from the supine position rather than the overstretched, weak, deep cervical flexors. To correct this muscle imbalance, begin training capital flexion as described in the stabilization section (core muscle activation). For home exercise, emphasize "curling" the head and neck, not lifting the head up.

Patient position and procedure: Supine. If the patient cannot tuck the chin and curl the neck to lift the head off the mat, begin with the patient on a slant board or large wedge-shaped bolster under the thorax and head to reduce the effects of gravity (Fig. 16.38). Have the patient practice tucking the chin and curling the head up. Use assistance until the correct pattern is learned. Progress by decreasing the angle of the board or wedge and then adding manual resistance if the patient does not substitute with the SCM.

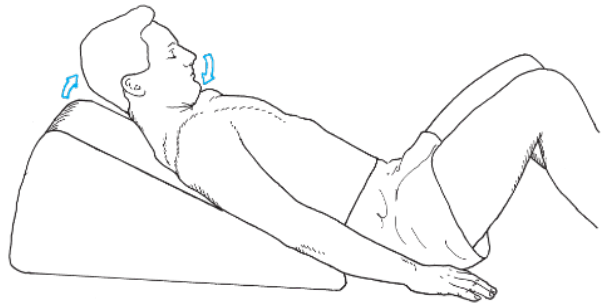


FIGURE 16.38 Training the short cervical flexors while de-emphasizing the sternocleidomastoid for cervical flexion to regain a balance in strength for anterior cervical stabilization.

Manual Resistance—Cervical Muscles

Patient position and procedures: Supine. Stand at the head end of the treatment table, supporting the patient's head for each exercise.

- Place one hand on the patient's head to resist opposite the motion. Do not resist against the mandible lest force be transmitted to the temporomandibular joint. Resistance is given to isolated muscle actions or to general ROMs, whichever best gains muscle balance and function.
- Isometric resistance can be applied with the head in any desired position before applying resistance. Avoid jerking the neck when applying or releasing the resistance by gradually building up the intensity, telling the patient to match your resistance, holding, and then gradually releasing and asking the patient to relax.

Intermediate and Advanced Training

As the patient progresses in the rehabilitation program, greater challenges to the musculature to stabilize and

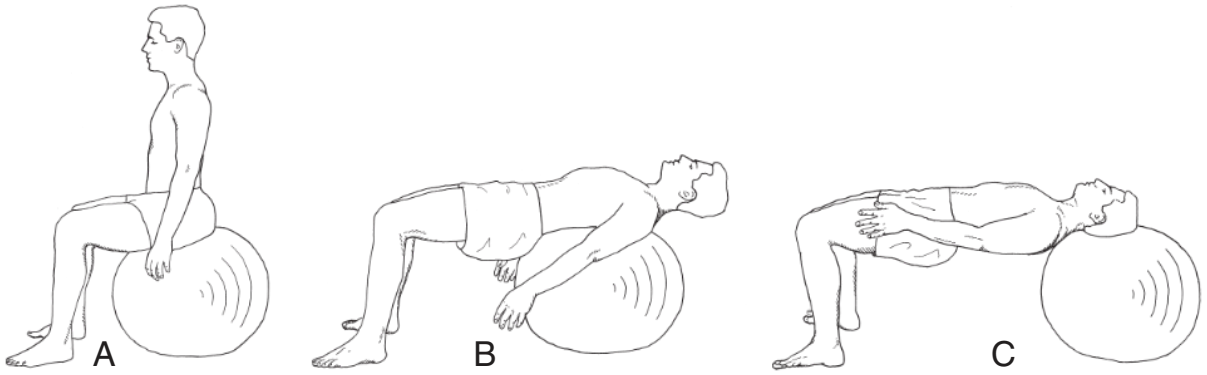


FIGURE 16.39 Advanced exercises for strengthening the cervical and upper thoracic flexors and extensors as stabilizers. Begin by (*A*) sitting on a large gym ball, then (*B*) walking forward while rolling the ball up the back. With the ball behind the mid-thoracic area, the cervical flexors must stabilize. Continue walking forward until the ball is (*C*) under the head; the cervical extensors now must stabilize. Walk back and forth between the two positions (*B* and *C*) to alternate control between the flexors and extensors. Progress by adding arm motions or arm motions with weights to increase resistance.

control motion are emphasized, especially for those individuals returning to work, sports, or recreational activities that place greater demands on the cervical structures.

Transitional Stabilization for the Cervical and Upper Thoracic Regions

- *Patient position and procedure:* Standing with a basketball-size inflatable ball between the head and the wall. Have the patient roll the ball along the wall, using the head. This requires the patient to turn the body as he or she walks along.
- *Patient position and procedure:* Sitting on a large gym ball. Have the patient walk the feet forward so the ball rolls up the back and the thorax is resting on the ball (Fig. 16.39 A&B). The head and neck are maintained in neutral position, and the cervical flexors are emphasized. Have the patient then walk the ball farther so it is under the head. The extensors are now emphasized (Fig. 16.39C). The patient walks the feet forward and backward, alternating stabilization between the flexors and extensors. Progress to advanced training by adding arm motions and then arm motions with weights in each of the positions.

NOTE: This activity requires considerable strength in the cervical extensors to support the body weight and should be performed only with advanced training with patients who have been properly progressed to tolerate the resistance.

Functional Exercises

Design exercises that simulate patient-specific functional activities. Identify what activities stress that individual's neck and have the patient practice modifications of those activities with the spine kept in neutral position. Include pushing, pulling, reaching, and lifting (see Functional Training section later in this chapter). Challenge the patient with increased repetitions and weight and in patterns that replicate functional demands.

Exercises for the Thoracic and Lumbar Regions

Alternating Isometric Contractions and Rhythmic Stabilization

Patient positions and procedures: Begin with the patient supine in the most stable position. Progress to sitting on a stable surface, sitting on an unstable surface such as a large gym ball, kneeling, and then standing. Sitting, kneeling, and standing require stabilizing action in the hip, knee, and ankle musculature, respectively, as well as the spinal muscles. Apply resistance directly against the patient's shoulders or pelvis, against a rod that is held by the patient (see Fig. 17.41), or against the patient's arms.

- Have the patient find the neutral spine position and then activate the stabilizing muscles with the drawing-in maneuver prior to applying the resistive force. Then instruct the patient to "meet my resistance" while applying a force to stimulate isometric contractions. Apply the resistance in alternating directions at a controlled speed while the patient learns to maintain a steady position.
- Initially, provide verbal cues, such as "hold against my resistance, but do not overpower me. Feel your abdominal muscles contracting. Now I'm pulling in the opposite direction. Match the resistance and feel your back muscles contracting."
- Progress by shifting the directions of resistance without the verbal cues and then by increasing the speed.
- Begin with alternating resistance in the sagittal plane; progress to side-to-side and then transverse plane resist-

ance. Isometric resistance to trunk rotation (transverse plane resistance) has been shown to be the most effective in stimulating the oblique abdominals, transversus abdominis, and deep spinal extensor muscles.[43]

- Alternating resistance to pelvic rotation can also be done by having the patient assume a modified bridge position. Apply resistance directly to the pelvis to stimulate rotation while the patient isometrically holds the pelvis and spine in a stable position.

Dynamic Strengthening—Abdominal Muscles

NOTE: Dynamic exercises of the trunk musculature are not initiated until late during the rehabilitation process and not until after the patient has learned to activate the drawing-in maneuver automatically for stabilization in all functional activities.

No one abdominal exercise challenges all of the abdominal muscles[33]; therefore, a variety of exercises should be included in the patient's exercise program to include the entire region.

Focus on Evidence

EMG studies have looked at abdominal muscle recruitment with various abdominal exercises.[3,29,33,54] In summary:

- Curl-ups (various types) recruit primarily the rectus abdominis, with low activity in the obliques, transversus abdominis, and psoas.
- Sit-ups (straight-leg and bent-knee) show high rectus and external oblique activity, high psoas activity, and high low-back compression. Heel press sit-ups increase psoas activity.
- Hanging-leg raises show high external oblique and high spinal compression.
- Supine single-leg lifts show negligible global abdominal muscle activity (opposite lower extremity provides stability). These exercises are primarily used early in the stabilization exercise routines to train the deep stabilizing muscles under progressive extremity loading.
- Supine bilateral-leg lifts show increased activity in the RA, EO, and IO during the first part of the range of hip flexion and increased load on the spine.
- Curl-ups on a labile surface doubled the activity of the rectus abdominis and increased the activity of the external obliques fourfold compared with curl-ups on a stable surface.[54]

Rectus abdominis. There is no clinically significant selective difference between the upper and lower rectus abdominis function.[29] Both portions contract strongly in all trunk curl-type and leg lift exercises.[29,33]
External obliques. External obliques contract strongest in sit-ups and diagonal sit-ups to the opposite side.[32]
Internal obliques. Internal obliques contract strongest in diagonal sit-ups to the same side and horizontal side propping (see Fig. 16.36).[33]

Trunk Flexion

Patient position: Supine or hook-lying with the lumbar spine neutral. McGill[33] suggested supporting the low back with the hands to maintain slight lordosis. The spine should not be allowed to go into an increased lordosis during the exercise—this indicates weakness of the abdominals and consequently lifting of the trunk occurs from hip flexor action only.[27] When training the abdominals, curl-up exercises should be performed at a slow, controlled rate to activate the stabilizing function of the abdominals.[55]

PRECAUTIONS: Patients with conditions such as intervertebral disk lesions or osteoporosis should not do dynamic trunk flexion exercises because of the increased intra-abdominal pressure and increased compression forces on the vertebral bodies. Similarly, if a patient experiences pain with trunk flexion, these exercises should not be done. Use the stabilization exercises, as described in the previous section, with the spine maintained in a neutral position (slight lordosis).

Procedures

Curl-ups. First, instruct the patient to perform the drawing-in maneuver to cause a stabilizing contraction of the abdominal muscles (see section on Stabilization Training—Core Muscle Activation) and then lift the head. Progress by lifting the shoulders until the scapulae and thorax clear the mat, keeping the arms horizontal (Fig. 16.40). A full sit-up is not necessary because once the thorax clears the mat the rest of the motion is performed by the hip flexor muscles.

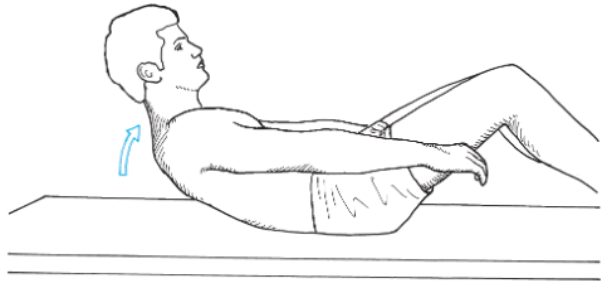


FIGURE 16.40 The curl-up exercise to strengthen the abdominal muscles. The thorax is flexed on the lumbar spine. The arms are shown in the position for least resistance. Progress by crossing the arms across the chest and then behind the head.

- Further progress the difficulty of the curl-up by having the patient change the arm position from horizontal to folded across the chest and then to behind the head.
- During all these activities, the low back should not arch; if it does, reduce the progression until the abdominals are strong enough to maintain lumbar flexion.

Curl-downs. If the patient is unable to perform the curl-up, begin with curl-downs by having the patient start in the hook-sitting or long-sitting position and lower the trunk only to the point where he or she can maintain a flat low back and then return to the sitting position.

- Once the patient can curl-down full range, reverse and perform a curl-up.

Diagonal curl-ups. Have the patient reach one hand toward the outside of the opposite knee while curling up; then alternate. Reverse the muscle action by bringing one knee up toward the opposite shoulder; then repeat with the other knee. Diagonal exercises emphasize the oblique muscles.

Double knee-to-chest. To emphasize the lower rectus abdominis and oblique muscles, have the patient set a posterior pelvic tilt; then bring both knees to the chest and return. Progress the difficulty by decreasing the angle of hip and knee flexion (Fig. 16.41).

Pelvic lifts. Have the patient begin with the hips at 90° and knees extended; then lift the buttocks upward off the mat (small motion). The feet move upward toward the ceiling (Fig. 16.42). The patient should not push against the mat with the hands.

Bilateral straight-leg raising

- Have the patient begin with legs extended; then perform a posterior pelvic tilt followed by flexing both hips, keeping the knees extended. If the pelvis and spine cannot be kept stable, the knees should be flexed to a degree that allows control.
- If the hips are abducted before initiating this exercise, greater stress is placed on the oblique abdominal muscles.

PRECAUTION: The strong pull of the psoas major causes shear forces on the lumbar vertebrae. Also this bilateral straight-leg raise (SLR) causes increased spinal compression loads. If there is any low-back pain or discomfort, especially with spinal hypermobility or instability, this exercise should not be performed even if the abdominals are strong enough to maintain a posterior pelvic tilt.

Bilateral straight-leg lowering. Bilateral straight-leg lowering can be performed if the bilateral SLR is difficult. Have the patient begin with the hips at 90° and knees extended; then lower the extremities as far as possible while maintaining stability in the lumbar spine (should not increase the lordosis), followed by raising the legs back to 90°. See precaution under the bilateral SLR exercise.

Trunk Flexion (Abdominals)—Sitting or Standing

Patient position and procedures: Sitting or standing. Pulleys or elastic material are secured at shoulder level behind the patient. Progress the resistance as the patient's abdominal strength increases.

- Have the patient hold the handles or ends of the elastic material with each hand and then flex the trunk, with emphasis on bringing the rib cage down toward the pubic bone and performing a posterior pelvic tilt, rather than flexing at the hips (Fig. 16.43).
- Have the patient perform diagonal motions by bringing one arm down toward the opposite knee with emphasis on moving the rib cage down toward the opposite side of the pelvis. Repeat the diagonal motion in the opposite direction.

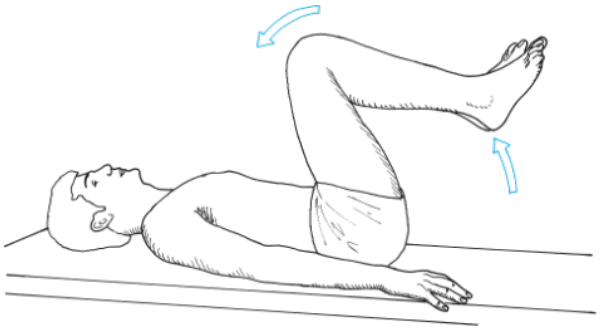


FIGURE 16.41 Strengthening the abdominal muscles by flexing the hip and pelvis on the lumbar spine. The legs are shown in the position for least resistance. Progress by decreasing the angle of hip flexion until the legs can be lifted with the knees extended, as in the pelvic lift.



FIGURE 16.42 Pelvic lifts. Elevating the legs upward toward the ceiling by raising the buttock off the floor emphasizes strengthening the lower abdominal muscles.



FIGURE 16.43 Standing trunk flexion against elastic material to strengthen the abdominal muscles. The patient performs a posterior pelvic tilt and approximates the ribs toward the pubis.

Trunk Flexion (Abdominals)—Unstable Surfaces

Patients with chronic, unilateral low back pain have been shown to have impaired balance.[2] Use of unstable surfaces, such as a gym ball (Fig. 16.44) or a balance board, while doing abdominal curl-up exercises has been shown to increase activity in the internal and external obliques and the rectus abdominis.[49] The presumption is that these muscles generate increased activity to maintain balance on the unstable surfaces. Other suggestions include balancing on a foam roll or a biomechanical ankle platform system (BAPS) board while performing arm and leg activities or curl-up exercises.

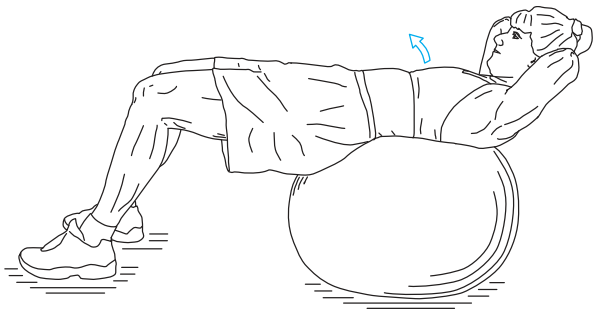


FIGURE 16.44 Curl-ups on an unstable surface. The unstable surface increases activity in the oblique and rectus abdominis muscles.

Dynamic Strengthening—Erector Spinae and Multifidus Muscles

Focus on Evidence

Strengthening the extensor muscles and an improved extensor/flexor ratio has been found to be important in decreasing symptoms in patients with chronic low back pain (LBP).[52] Lee et al.[28] determined that the trunk extensor/flexor ratio is a sensitive parameter for predicting LBP. After following 67 asymptomatic individuals for 5 years, they found an increased incidence of LBP in those who had lower extensor than flexor muscle strength. Danneels et al.[9] demonstrated that intensive lumbar resistance training (isometric or dynamic) is necessary to develop paravertebral muscle strength and bulk. The following is a summary of specific exercise outcomes studies.

- Prone arch, isometric trunk extension, and isometric leg extensions: high activity in both the multifidus and erector spinae.[36]
- Isolated training of multifidus: requires a low-intensity focus, as described in the stabilization section—techniques for core muscle activation.[40]

Resistance can be applied to any of the following recumbent exercises by having the patient hold weights in the hands or by strapping weights around the patient's legs.

Extension Exercises in Prone Position

PRECAUTIONS: Extension exercises in the prone position are performed at the end of the ROM in spinal extension and therefore may not be appropriate for individuals with symptoms from conditions such as arthritis or nerve root compression. Patients with spondylosis or other flexion bias conditions or patients who develop symptoms under loaded conditions (e.g., with disk lesions) may experience increased symptoms and therefore should not do dynamic end-range extension exercises. If symptoms occur, modify the positioning toward more neutral spinal positions, such as the quadruped position, and emphasize stabilization with isometric holds rather than moving into full extension (see Figs. 16.28, 16.29, and 16.33).

Thoracic elevation. Begin with the arms at the side, progress to behind the head or reaching overhead as strength improves. Have the patient tuck in the chin and lift the head and thorax. The lower extremities must be stabilized (Fig. 16.45).

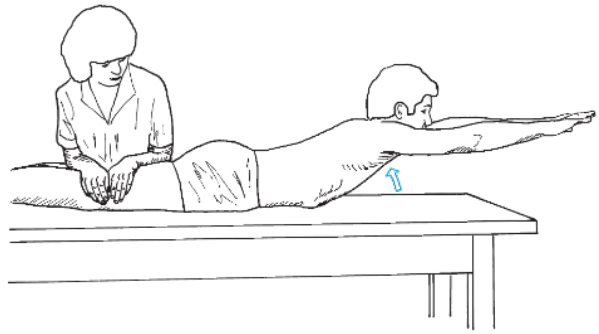


FIGURE 16.45 Strengthening the back extensors with the arms in position to provide maximal resistance. Additional resistance can be provided by holding weights in the hands.

Leg lifts. Initially have the patient lift only one leg, alternate with the other leg, and finally lift both legs and extend the spine. Stabilize the thorax by having the patient hold onto the side of the treatment table.

"Superman." Progress the extension exercises by having the patient lift both upper and lower extremities simultaneously (Fig. 16.46).

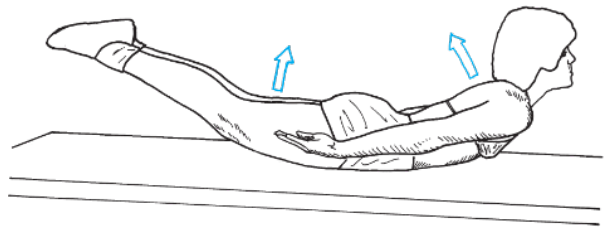


FIGURE 16.46 Strengthen the trunk and hip extensors by lifting the trunk and legs off the mat simultaneously. Greater resistance can be provided by abducting the shoulders to 90° or by elevating them to 180° ("Superman").

Extension Exercises Sitting or Standing

Elastic resistance or weighted pulleys. Secure pulleys or elastic resistance in front of the patient at shoulder level. Have him or her hold onto the ends of the material or handles and extend the spine (Fig. 16.47).



FIGURE 16.47 Using elastic resistance for concentric eccentric back extension.

For trunk rotation, use a pulley or elastic resistance secured under the foot or to a stable object opposite to the side being exercised. Have the patient pull against the resistance, extending and rotating the back. Change the angle of pull of the resistance to recreate functional patterns specific to the patient's needs (Fig. 16.48).



FIGURE 16.48 Rotation with extension strengthens the back extensors in functional patterns.

Trunk Side Bending (Lateral Abdominals, Erector Spinae, Quadratus Lumborum)

Trunk side-bending exercises are used for general strengthening of the muscles that side-bend the trunk.

Focus on Evidence

McGill[33] identified the quadratus lumborum as one of the most important stabilizers of the spine and documented the isometric horizontal side support as an effective exercise to strengthen this muscle (see discussion in the stabilization section and Figure 16.36).

Side-bending exercises are also used if there is scoliosis, although exercise alone has not been shown to halt or change the progression of a structural scoliosis curve. Exercise in conjunction with other methods of correction, such as bracing, is often employed.[5,7] When there is a lateral curve, the muscles on the convex side are usually stretched and weakened. The following exercises are described for use as strengthening exercises on the side of the convexity, although they may be used bilaterally for symmetrical strengthening.

Stabilization exercises for spinal control, as previously described, may be beneficial for strengthening and conditioning when there is scoliosis.

- *Patient position and procedure:* Standing. Place elastic resistance under the foot or have the patient hold a weight in the hand on the side of the concavity; then have him or her side-bend the trunk in the opposite direction.
- *Patient position and procedure:* Side-lying on the concave side of the curve with the apex at the edge of the table or mat so the thorax is lowered. If you have access to a split table with one end that can be lowered, begin with the apex of the curve at the bend of the table. Have the patient place the lower arm folded across the chest and upper arm along the side of the body and side-bend the trunk up against gravity. Progress by having the patient clasp both hands behind the head (Fig. 16.49). Stabilization of the pelvis and lower extremities must be provided.

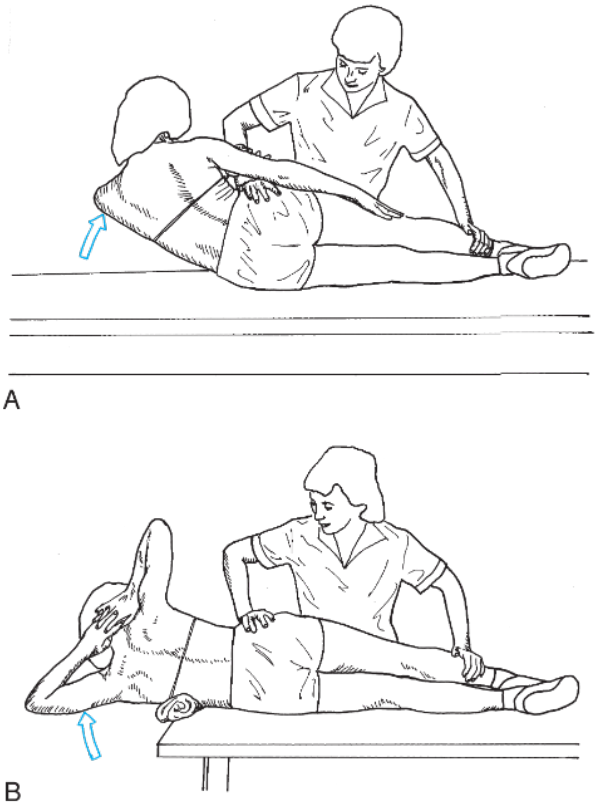


FIGURE 16.49 (A, B) Antigravity strengthening of the lateral trunk musculature. There is less resistance if the top arm is at the side and the bottom arm is folded across the chest.

CARDIOPULMONARY ENDURANCE

Goal. To develop cardiopulmonary fitness for overall endurance and well-being.

Aerobic conditioning exercises provide many benefits for the patient with spinal symptoms. The activity not only improves cardiopulmonary endurance but stimulates feelings of well-being and relief of symptoms.[35] Chapter 7 describes cardiopulmonary conditioning principles and procedures. Specific precautions and suggestions for medical conditions are also explained. For patients recovering from spinal injuries, surgery, or postural dysfunction, aerobic exercises may be initiated once signs of inflammation no longer exist. Begin with low to moderate intensity and work with the patient to choose activities that do not place added stress on the recovering spinal structures. If a particular spinal bias has been identified (see Chapter 15), choose aerobic exercises that emphasize that spinal bias. A brief summary of the principles is reviewed in Box 16.6. Guidelines for safe application of common conditioning exercises when there are spinal impairments are described in this section.

BOX 16.6 Summary of Aerobic Conditioning Principles

- Establish the target heart rate and maximum heart rate.
  - The maximum heart rate is generally 220 minus the individual's age or may be the symptom-limiting heart rate (the rate at which cardiovascular symptoms appear).
  - Target heart rate is between 60% and 80% of the maximum heart rate.
- Perform warm-up exercises for 10 to 15 minutes, including active movements of the neck and trunk.
- Individualize the program of exercise.
  - Select activities that emphasize the patient's spinal bias if necessary (see information in the text).
  - Not all people are at the same fitness level and therefore cannot perform the same exercises. Any one exercise has the potential to be detrimental if attempted by someone not able to execute it properly.
  - To avoid overuse syndromes to structures of the musculoskeletal system, appropriate equipment, such as correct footwear, should be used for biomechanical support with weight-bearing exercises.
- Increase the pace of the activity to reach the target heart rate and maintain it for 20 to 30 minutes.
- Cool down for 5 to 10 minutes with slow, total body, repetitive motions and stretching activities.
- Frequency of aerobic exercise should be three to five times per week.
- Always stay within the tolerance of the individual. Overuse commonly occurs when there is an increase in time or effort without adequate rest (recovery) time between sessions. Increase repetitions or time by no more than 10% per week.[30] If pain begins while exercising, heed the warning and reduce the stress.

COMMON AEROBIC EXERCISES AND EFFECTS ON THE SPINE

Some aerobic exercises place the spine in end-range positions. They are reviewed so the reader understands why some activities may be inappropriate for patients with specific conditions. If modifications are possible, they should be considered.

Cycling

Road bikes place the thoracolumbar spine in flexion and the upper cervical spine in hyperextension. Use this exercise for patients who have a flexion bias in the lumbar region so long as there are no upper cervical symptoms. Modifications include using a bike that positions the body in a more upright posture, such as a mountain bike or

hybrid bike. Many stationary bikes also position the individual in upright postures and therefore are less likely to precipitate cervical problems.

Walking and Running

The upright posture emphasizes normal spinal curves, and lumbar extension is emphasized with walking and running (terminal stance). Emphasize the importance of identifying the neutral spine, activating the drawing-in maneuver, and stabilizing the spine while walking or running. Because conscious control is not possible during the entire exercise time, coach the patient to check his or her posture and muscle control frequently, such as each time he or she is crossing an intersection, or when passing another individual, or if symptoms develop in the spinal region. Walking or running with the cervical spine in retraction (axial extension) and the scapulae comfortably adducted, along with a rhythmic arm swing, reinforces cervical stabilization. Easy access to treadmills, tracks, or roads and trails make these activities popular. Running is a high-impact activity and may not be tolerated by individuals with intervertebral disk lesions or degenerative joint conditions.

Stair Climbing

Commercial devices that replicate stepping with various grades of resistance are used for strengthening and aerobic conditioning. Regular steps can also be used for aerobic conditioning. This activity requires pelvic control of the reciprocating lower extremities because lifting the leg on one side emphasizes spinal flexion while the contralateral lower extremity and spine are extending. Coach the patient to maintain the neutral spine with the stabilizing muscles against the rotational forces.

Cross-Country Skiing and Ski Machines

Cross-country skiing, whether out in the cold or on a commercial machine, is a high-intensity aerobic activity. The kicking motion that accompanies the backward motion of the leg emphasizes spinal extension. It is important to coach the patient to maintain the neutral spine and contract the stabilizing abdominal muscles.

Swimming

When swimming, the breaststroke emphasizes extension in the cervical and lumbar spinal regions when taking a breath. Coach the patient to not extend the neck full range but to keep it neutral and lift the head out of the water as a "solid" unit with the thorax just enough to clear the mouth for the breath.

The freestyle stroke may exacerbate cervical problems because of the repetitive cervical rotation while taking a breath; this stroke also emphasizes lumbar extension with the flutter kick. Teach the patient to breathe using a "log-roll" technique where the whole body rolls toward one side while breathing and then rolls back to the face-down position for the stroke. This requires good spinal stabilization.

The backstroke emphasizes spinal extension via kicking the lower extremities and the arm motions. The butterfly stroke moves the spine through a full ROM; emphasis is placed on controlling the range with the stabilizing muscles.

Upper Body Ergometry Machines

Ergometry machines provide upper extremity resistance and can also be used for aerobic training. Forward motions emphasize spinal flexion and shoulder girdle protraction; backward motions emphasize spinal extension and shoulder girdle retraction. Coach the patient to assume the neutral spinal posture and use the stabilizing muscles prior to and during the use of the ergometer to enhance postural responses. If the machine can be used standing, progression to the standing position stimulates a total body response.

Step Aerobics and Aerobic Dancing

Stepping is similar to using stairs or a stair machine except for the jumping and bouncing that is usually added to the more advanced step aerobic programs.

Dancing moves take on many forms, and classes are taught that address various fitness levels and age groups. If possible, review safe movement patterns and help the patient recognize the safe limits of his or her spinal range and abilities.

"Latest Popular Craze"

People like variety and may be attracted to charismatic and energetic figures who demonstrate "new" workout techniques and routines or new exercise machines. Patients may ask for advice as to the value of the activities and techniques. Knowledge and skill in analyzing the biomechanics of the activity and the forces that are imposed through the spine should be used to provide advice about exercise safety. End-of-range postures and high-velocity stresses (such as vigorous kicking and ballistic motions) may be damaging to vulnerable tissues in the spine and should not be attempted by patients recovering from spinal problems.

FUNCTIONAL ACTIVITIES

Goal. To progress to independence safely.

NOTE: Achieving the maximum level of independence underlies all the goals of therapeutic exercise. The patient develops core and global stability; develops flexibility, muscle endurance, and strength; learns how exercise and posture correction relieves stress; and develops cardiopul-

monary endurance—all to be able to function safely in daily activities, including work, recreation, and athletic pursuits.

Early Functional Training—Fundamental Techniques

Early functional training consists of teaching basic maneuvers needed for ADLs such as how to roll over safely, go from lying down to sitting (and reverse), and go from sitting to standing (and reverse). These techniques follow the early kinesthetic training instruction in which the patient learns to find his or her neutral spine and experiences the effect that simple arm and leg motions have on the spine, as well as early muscle performance training in which the patient learns how to activate the core musculature for segmental stabilization. If the examination reveals problems with basic ADL activities, the following are included in the early training program.

Rolling. Rolling with a neutral spine requires that the patient first find the neutral spine, perform the drawing-in maneuver, and then roll the trunk as a unit.

- It may be helpful to suggest that the patient "imagine a solid rod connecting the shoulders and pelvis so as not to twist" or suggest that he or she "roll like a log."
- Encourage the patient to use the arms and top leg to assist the roll.

Supine to sit/sit to lying down. Have the patient use the log roll maneuver (as described above) to roll from supine to side-lying while simultaneously flexing the hips and knees and pushing up with the arms.

- Help the patient focus on stabilizing the trunk with commands such as "push up your trunk as if it is a board; do not allow it to twist or bend."
- The reverse is practiced by coaching the patient to lower oneself to the side-lying position as a unit first onto the elbow and then shoulder. Once down, the patient can roll to supine or prone-lying using the log-roll technique

Sit to stand/stand to sit. The patient's level of function dictates how much assistance from the upper extremities is needed to accomplish "sit to stand" or "stand to sit." If the hip and knee extensors are not strong enough to elevate the body, the patient requires a chair with armrests so there is some leverage for pushing up; alternatively, an elevated seat may be necessary.

- To use the stable spine technique, instruct the patient to find the neutral spine by rolling the pelvis forward and backward, activate the drawing-in maneuver, and then bend forward at the hips while maintaining the neutral spine position.
- Help the patient focus on the hip motion while keeping the spine "solid like a board." The reverse is also practiced.

In and out of a car. Getting in and out of a car is often symptom-provoking for patients with low back pain. Once sit to stand can be safely performed, have the patient practice the following.

- Approach the open car door and seat with the back toward the seat; stabilize the spine in its neutral position with the drawing-in maneuver, then bend at the hips and sit down.
- Once seated, flex both hips and knees and pivot the whole body around as a unit, maintaining a stable spine.
- When exiting a car, keep both knees together and pivot the legs and trunk outward as a unit. Once the feet are on the ground, bend at the hips and elevate the trunk as a unit.

Walking. For some patients walking may provoke symptoms.

- Remind the patient to use the neutral spine and drawing-in maneuvers to stabilize the spine while walking.
- It is not possible to maintain conscious control for long, so remind the patient to check the spinal posture and reactivate the drawing-in maneuver whenever the symptoms recur.

Preparation for Functional Activities—Basic Exercise Techniques

Once the patient has learned to manage his or her symptoms and the symptoms of inflammation diminish, exercises are initiated that prepare the extremities and trunk for functional activities such as safely lifting, carrying, pushing, pulling, and reaching in various directions. In the subacute or controlled motion phase of rehabilitation, emphasis is placed on strengthening the extremities in functional patterns while maintaining a stable spine. The patient should be able to perform IADLs and limited work activities at this stage. Evaluate the patient's performance and modify what he or she is doing to include safe spinal postures and correct stabilization. Use the activities in this section to prepare for or advance the patient's function.

Many of the strengthening exercises described in the extremity chapters are appropriate to use in preparation for functional training. With postural problems and recovery from back or neck injuries, it is critical to emphasize the neutral (functional) spinal posture before and during total body exercises. Many of the stabilization and movement patterns described earlier in this chapter can also be progressed in intensity, repetitions, speed, and coordination to prepare for return to functional activities.

Weight-Bearing Exercises

Modified Bridging Exercises

Modified bridging exercises require stabilization with the trunk flexor and extensor muscles in conjunction with strengthening the gluteus maximus and quadriceps muscles in preparation for lifting activities. The abdominals function with the gluteus maximus to control the pelvic tilt, and

the lumbar extensors stabilize the spine against the pull of the gluteus maximus.

Patient position and procedures: Begin with the patient hook-lying. Have the patient concentrate on maintaining the neutral spinal position while raising and lowering the pelvis (flexing and extending at the hips) (see Fig. 20.21). Hold the bridge for isometric control.

- Alternate arm motions; progress by adding weights to the hands.
- Alternate lifting one foot and then the other by marching in place (Fig. 16.50A); progress by extending the knee as each leg is lifted. When the patient tolerates greater resistance, add ankle weights and coordinate with arm motions (Fig. 16.50B).
- Abduct and adduct the thighs without letting the pelvis sag. Progress by placing the feet on a stool, chair, or large gym ball and repeating the bridging activities.

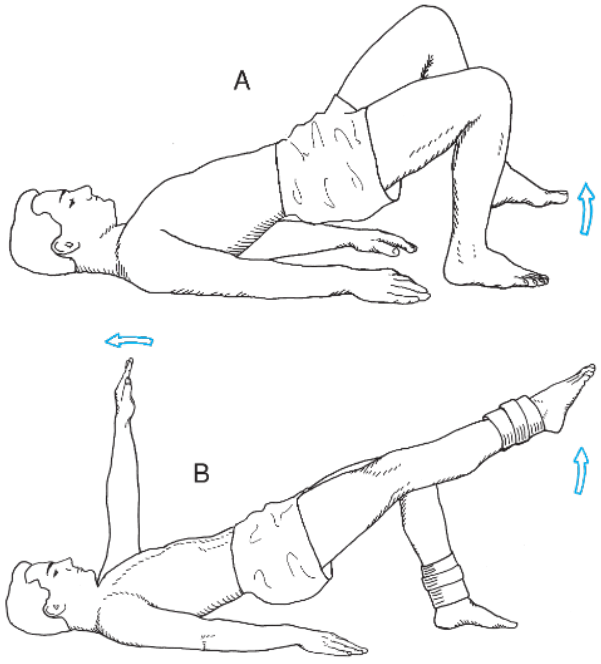


FIGURE 16.50 Holding a bridge to develop trunk control and gluteus maximus strength while superimposing extremity motions by (*A*) marching in place and (*B*) progressing to extending the extremities. Adding weights to the arms or legs requires greater strength and control.

Push-Ups with Trunk Stabilization

Push-ups use the body weight to strengthen the triceps and shoulder girdle musculature in preparation for pushing activities. The trunk musculature must stabilize against the pull of the shoulder girdle musculature as well as control the neutral spinal position as the body is raised and lowered.

Patient positions and procedures: Standing facing a wall or prone-lying with hands placed against the wall or floor in front of the shoulders. Remind the patient to find and maintain the neutral spinal position while performing the exercise.

- These exercises may begin as wall push-ups if the patient is not strong enough to push up from the floor.
- Prone-lying on the floor the patient may push up with the pivot point being the knees or may perform full body push-ups with the pivot point being the feet.
- To challenge the patient on an unstable surface, he or she begins prone on a large gym ball. Have the patient walk forward with the hands on the floor until just the thighs are supported by the ball, maintain a stable spinal posture, and perform push-ups with the arms. To progress, walk out farther with the hands until just the legs are supported by the ball (Fig. 16.51).

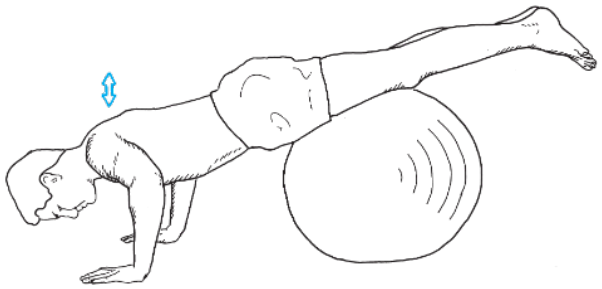


FIGURE 16.51 Push-up activities with the lower extremities balanced on a gym ball for strengthening the arms and developing trunk control.

Wall Slides

Wall slides develop strength in the hip and knee extensor muscles to prepare the lower extremities for squatting activities and training in safe body mechanics.

Patient position and procedure: Standing with the back to a wall and the spine held in its neutral position. Place a towel behind the back so it slides easier along the wall. The exercise is more challenging if a large gym ball is placed between the back and the wall (Fig. 16.52). Have

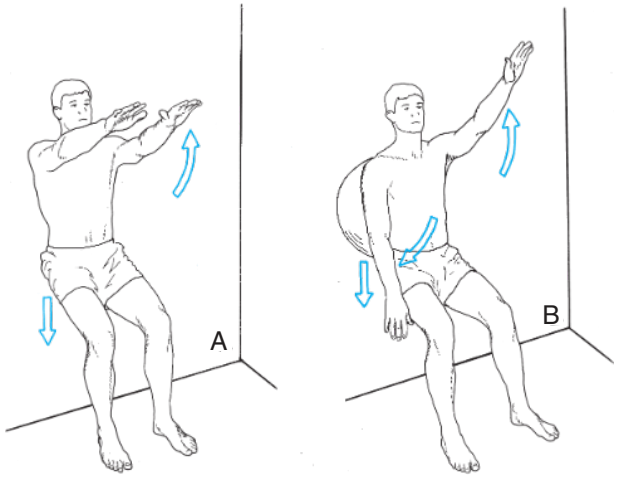


FIGURE 16.52 Wall slides/partial squats to develop lower extremity strength and coordinate with trunk stability in preparation for training body mechanics. (*A*) The back sliding down a wall, with bilateral arm motion for added resistance. (*B*) Rolling a gym ball down the wall, with antagonistic arm motion to develop coordination.

the patient slide his or her back down the wall into a partial squat and hold the position for isometric strengthening of the hip and knee extensors or move up and down for concentric/eccentric strengthening.

- Superimpose arm motions such as alternating or bilateral shoulder flexion/extension.
- Use handheld weights to add resistance for upper and lower extremity strengthening.

Partial Lunges and Partial Squats

Partial lunges and squats are described in Chapters 20 and 21. They are beneficial for strengthening total body movement in preparation for learning body mechanics. If necessary, begin by having the patient balance by holding onto the side of a treatment table or other stable object and then progress to balancing with a cane (see Fig. 20.23). Once able to perform multiple repetitions without holding on, add weights to the upper extremities for resistance. Progress by adding arm motions that are synchronized with the leg motions, such as reaching forward and downward to develop coordination and control.

Walking Against Resistance

Secure a weighted pulley or elastic resistance around the patient's pelvis with a belt, or the patient can hold the handles. Have the patient walk forward, backward, or diagonally against the resistive force. Emphasis is placed on spinal control (see Fig. 20.25).

Progress by having the patient push and pull weighted objects, such as a cart or a box on a table. Place emphasis on maintaining a stable spinal position while the extremities are loaded (see Figs. 17.56 and 18.20A).

Transitional Stabilization Exercises

Exercises that cause movement into spinal flexion and extension challenge the patient to control the neutral spine position. The patient learns to stabilize the spine against alternating trunk motions.

Quadruped Forward/Backward Shifting

Patient position and procedure: Quadruped. Have the patient rock back to rest the buttocks on the heels, then shift the body forward onto the hands in the press-up position. The patient concentrates on controlling the pelvis in its neutral position rather than allowing full spinal flexion when shifting toward the heels or full spinal extension when shifting forward toward the press-up position.

Squatting and Reaching

Patient position and procedure: Begin standing. Have the patient reach downward while partially squatting. The tendency is for the spine to flex, so have the patient concentrate on maintaining a neutral spinal position with the spinal extensors. The patient then stands up and reaches overhead. This causes the spine to extend, so have the patient concentrate on using the trunk flexors to stabilize in the neutral position. Progress by lifting and reaching with weights while controlling the neutral posture of the spine.

Shifting Weight and Turning

Have the patient practice shifting weight forward/backward and side to side while maintaining the neutral spinal position and absorbing the forces with the hip and knee muscles. Practice turning using small steps and rotating at the hips rather than the back. Instruct the patient to imagine two rigid poles connecting each shoulder to each hip that do not allow the spine to twist. Even though some movement in the spine occurs, the activity helps the patient focus on a stable spine rather than rotating full range. Progress by using weights and having the patient lift, turn, and then place the weight at a new location.

Body Mechanics and Environmental Adaptations

Principles of Body Mechanics—Instruction and Training

When teaching safe body mechanics it is advisable to not overwhelm the patient with too many instructions. Most people "know" they are to lift with their legs rather than their back, but still have faulty techniques. Initiate training by suggesting that the patient find his or her neutral spine, perform the drawing-in maneuver then lift. Observe the technique they use and suggest modifications if needed. Squatting is often taught as the preferred method, yet not all patients are able to squat if they have impairments such as knee pain or weakness. Under some circumstances, an individual may be more stable lifting with a lunge technique rather than the squat technique.

Lumbar Spine Position

The position of the lumbar spine, whether it is flexed, extended, or in mid-range, raises several issues. Of the three postures, lifting with a neutral spinal posture provides greater stability of the spine[12] and uses both the ligamentous and muscular system for stabilization and control.[50] After a back injury, the preferred lifting posture may have to be adapted, depending on the type of injury and the response of the tissues when stressed.[50]

Spinal flexion. When lifting with a flexed lumbar spine (posterior pelvic tilt), support for the spine is primarily from inert structures (ligaments, lumbodorsal fasciae, posterior annulus fibrosus, and facets); there is little muscle activity.

- Flexion occurs when stooping to the floor. Some have suggested that it may also be the posture of choice for a patient who has injured the back muscles because the muscles are "quiet" when the spine is in flexion.[50]
- Lifting with the lumbar spine in flexion may pose some problems. When lifting slowly with a flexed spine, the load is maintained on the ligaments, and creep of the

inert tissues occurs; this increases the chance of injury if the tissue is already weakened. In addition, with the muscles lengthened and relaxed, they may be at an unfavorable length–tension relationship to respond quickly with appropriate force to resist a sudden change in load. There is greater chance of ligamentous strain when a person lifts with a flexed spine.[22]

Spinal extension. When lifting with an extended (lordotic) lumbar spine, the muscles supporting the spine are more active than when flexed, which increases the compressive forces on the disk. Also, the facets are approximated (close-pack position). This posture relieves stress on the ligaments; but for an individual whose back muscles are in poor condition and fatigue quickly, this posture may jeopardize the spine when repeated lifts are performed because the ligaments are not providing support.[32]

Load Position

Reinforce the concept of lifting and carrying objects as close to the center of gravity as possible.

- Have the patient practice carrying objects close to his or her center of gravity and draw attention to the feel of balance and control, as well as less stress on the neck and back, compared to the feel when carrying objects in more stressful positions. Point out that, when lifting, the closer the object is held to the center of gravity, the less stress is placed on the supporting structures.
- Have the patient practice shifting the load from side to side and turning. Have the patient practice turning with hip rotation and minimal trunk rotation. The action should be directed by the legs while the spine is kept stable.
- Replicate the mechanics of the patient's job setting and practice safe mechanics.

Environmental Adaptations

Ergonomic assessment and modification of the home and working environments are necessary to correct stresses as well as prevent future recurrence of symptoms.

Home, Work, and Driving Considerations

- Chairs and car seats should have lumbar support to maintain slight lordosis. Use a towel roll or lumbar pillow if necessary.
- Chair height should allow knees to flex to take tension off the hamstring muscles, support the thighs, and allow the feet to rest comfortably on the floor.
- Arm rests should be used if prolonged sitting is required in order to take the stress off shoulders and the cervical spine.
- Desk or table height should be adequate to keep the person from having to lean over the work.
- Work and driving habits should allow frequent changing of posture. If normally sedentary, the patient should get up and walk every hour.

Sleeping Environment

- The mattress needs to provide firm support to prevent any extreme stresses. If it is too soft, the patient sags and stresses ligaments; if it is too firm, some patients cannot relax.
- Pillows should be of a comfortable height and density to promote relaxation but should not place joints in an extreme position. Foam rubber pillows tend to cause increased tension in muscles because of the constant resistance they provide.
- Whether the person should sleep prone, side-lying, or supine is something that must be analyzed for each individual patient. Ideally, a comfortable posture is one that is in the mid-range and that does not place stress on any supporting structure. Pain that is experienced when waking up in the morning is often related to sleeping posture; if this is the case, listen carefully to the patient's description of postures when sleeping and see if it relates to the pain. Then attempt to modify the sleep position accordingly. Remind the patient that it takes several weeks to change habits.

Intermediate to Advanced Exercise Techniques for Functional Training

As the patient learns spinal control while doing the exercises, repetitions are increased to develop muscular endurance, and resistance is added to develop strength. If speed and balance are required, they are emphasized. By this stage it is recognized that the individual already knows the basic spinal stabilization techniques and is habitually assuming the neutral spinal position and activating the drawing-in maneuver. Reinforce the importance of this when doing the following exercises. It is also recognized that the patient should be able to control greater spinal ROM without experiencing symptoms. Adapt the exercises to replicate return to work or sport-related activities. Examples follow.

Repetitive Lifting

The ability to do repetitive lifting throughout the workday is necessary for many jobs and may result in symptom recurrence. To prepare for returning to work, progressively increase the repetitions of lifting activities the patient must do to improve muscle endurance. Marras and Granta[31] demonstrated that with repetitive lifting (over a 5-hour period) subjects had a significant change in their lifting pattern and in the muscle recruitment patterns so there was a decrease in spine stabilization (decreased compression) and an increase in anterior/posterior shear in the lumbar spine. To reduce the risk of recurrence of low-back disorders, a patient needs to learn to monitor these changes and

be conscious of correcting faulty patterns. Help the patient modify and adapt the stable spine body mechanics that were initiated under basic techniques to replicate the type of lifting he or she will be doing at home or on the job. Include variations in the lifting tasks to prepare for unexpected situations.

Repetitive Reaching

Repetitive reaching requires that the patient learn to assume a comfortable stride and then practice shifting his or her weight forward and backward on the lower extremities rather than bending forward and backward with the spine. Preparatory exercises should include partial lunging forward, sideways, and backward. During practice have the patient use a weight comparable to that in the real-life situation and go through the action on a repetitive basis, concentrating on spinal control and resting only when control is no longer possible.

Repetitive Pushing and Pulling

Repetitive pushing and pulling require strong upper extremities and a stable spine. Preparatory activities should include pushing and pulling against elastic resistance or pulley resistance set at heights that replicate the work environment. Progress to pushing and pulling a weighted cart or a weighted box across a table. Reinforce the importance of activating the spinal stabilizers.

Rotation or Turning

Turning with a load is a component of most work activity. A person may rotate the spine to reach around to place a load to the side or behind. Rotation may create an unstable situation or may be damaging to the spinal structures. Therefore, it is important to take the rotation out of turning. Have the patient practice a "stable spine turn," which requires motion and control in the hips, or taking steps into the direction of the turn rather than twisting and rotating the back.

Transitional Movements

Most functional activities require *transitional motions*, such as reaching downward to pick up something (spinal flexion), then reaching overhead to place it on a high shelf (spinal extension). In sports activities the activity may require moving quickly from a forward-bent position to an extended position with arms overhead (such as dribbling a basketball, then shooting). Set up drills that replicate the speed and movements of the desired outcome; have the patient practice moving through the patterns while attempting to maintain control of his or her functional spinal position and range.

Transfer of Training

Ideally, each patient is progressed through rehabilitation to the level of being able to *transfer skills* learned to closely related but new situations. Provide variable learning opportunities from simple to complex and then help the patient analyze successful adaptations to each new experience. (See Figure 1.6 and accompanying text in Chapter 1 for examples of how to vary tasks from simple to complex.)

Education for Prevention

Education occurs on a continual basis. Before discharge, review the following relationships of posture and pain with the patient.

- When experiencing pain or the recurrence of symptoms, check posture. Avoid any one posture for prolonged periods.
- If sustained postures are necessary, take frequent breaks and perform appropriate ROM exercises at least every half hour. Finish all exercises by assuming a well balanced posture.
- Avoid hyperextending the neck or being in a forward-head posture or forward-bent position for prolonged periods. Find ways to modify a task so it can be accomplished at eye level or with proper lumbar support.
- If in a tension-producing situation, perform conscious relaxation exercises.
- Use common sense and follow good safety habits.

Develop and progress a safe home exercise program.

- Teach flexibility, muscle endurance, and strengthening exercises appropriate for the patient to maintain ROM, muscle endurance, and strength.
- Address any misconceptions the patient may have about exercise.

Reinforce the importance of maintaining cardiopulmonary endurance and its effect on managing symptoms.

INDEPENDENT LEARNING ACTIVITIES

Critical Thinking and Discussion

1. Observe a homemaker or worker doing an activity that requires pushing, pulling, reaching, lifting, or some other repetitive pattern. Analyze what component motions are part of the total pattern and decide if strength, range, endurance, balance, or coordination (or a combination) is necessary in the upper extremities, lower extremities, and trunk. Decide what is necessary to make the spine safe while doing this activity and design an exercise program that encompasses all the components.
2. Go to a health club or exercise class and observe how individuals are performing the exercises. Note the activities that cause stress to the spine. How would you modify each exercise? Consider safe use of the equipment, safe biomechanics, and appropriate instruction for the audience. Can you tell the purpose of each exercise (strength, stretch, endurance)? Are the directions appropriately given for the level of participants?

Laboratory Practice

1. With a laboratory partner practice the kinesthetic training techniques and core muscle activation techniques for the cervical spine and the lumbar spine until you become proficient at performing them and recognizing when they are done correctly. Then practice teaching them to a family member or friend and see how well they understand what they are to do.
2. Practice the progression of spinal stabilization exercises described in the muscle performance section. Start at the easiest level and progress the leg and arm movements until you feel you are at your maximum resistance for stabilization. After resting, time yourself for 1 minute, beginning at the most difficult level of movement. The idea is to keep the spine stable during the entire minute. If you begin to feel you are losing control, decrease the amount of extremity resistance (e.g., going from moving both lower extremities in a reciprocal pattern to moving just one extremity while the other is on the floor). This can also be done for 3 minutes. Were you able to meet the challenge yet keep the spine stable? Did you feel your stabilizing muscles "working?"
3. Practice doing wall slides, partial squats, and partial lunges with a stable spine. When you can do the squat comfortably with a stable spine, practice lifting a box from the floor to table height, then from the floor to shoulder height, then place it on a shelf at each height. Feel what is happening to your spine. Then repeat the maneuvers with a stable spine, and see if you can control the spinal position with the drawing-in maneuver. When you can do the lunge comfortably, practice lifting small objects from the floor with a lunging technique and stable spine. Finally, practice lifting objects from the floor and turning (using legs and hips to change direction, not spinal rotation) to place the objects on a table or shelf. Feel what is happening to the spine and repeat the activities with a stable spinal posture.

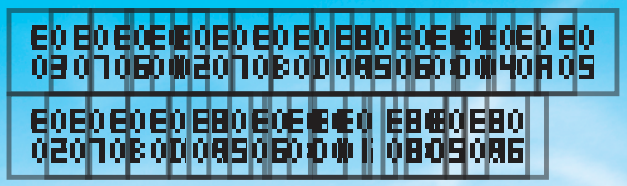
Case Studies

Review the cases described in Chapter 14 and 15 and modify your answers based on information you studied in this chapter.

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The design of the shoulder girdle allows for mobility of the upper extremity. As a result, the hand can be placed almost anywhere within a sphere of movement, being limited primarily by the length of the arm and the space taken up by the body. The combined mechanics of its joints and muscles provide for and control the mobility. When establishing a therapeutic exercise program for impaired function of the shoulder region, as with any other region of the body, the unique anatomical and kinesiologic features must be taken into consideration as well as the state of pathology and functional limitations imposed by the impairments. This chapter is divided into three major sections. The first section briefly reviews the structure and function of the shoulder girdle complex. The second section describes common disorders and guidelines for conservative and postsurgical management. The last section describes exercise techniques commonly used to meet the goals of treatment during the stages of tissue healing and phases of rehabilitation.

STRUCTURE AND FUNCTION OF THE SHOULDER GIRDLE

The shoulder girdle has only one bony attachment to the axial skeleton (Fig. 17.1). The clavicle articulates with the sternum via the small sternoclavicular joint. As a result, considerable mobility is allowed in the upper extremity. Stability is provided by an intricate balance between the scapular and glenohumeral muscles and the structures of the joints in the shoulder girdle.

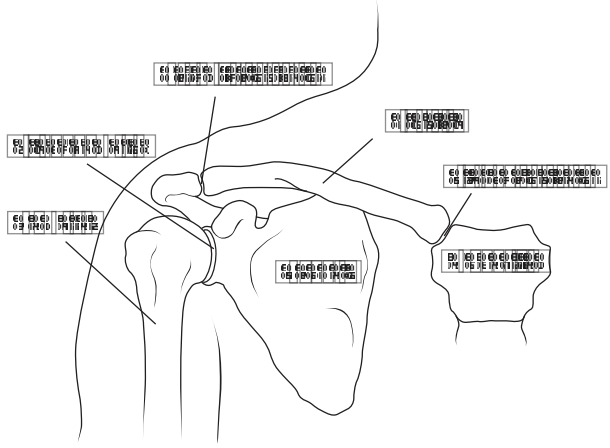


FIGURE 17.1 Bones and joints of the shoulder girdle complex.

JOINTS OF THE SHOULDER GIRDLE COMPLEX

There are three synovial joints (glenohumeral, acromioclavicular, sternoclavicular) and two functional articulations (scapulothoracic, suprahumeral) that make up the shoulder girdle complex.

Synovial Joints

Glenohumeral Joint

The glenohumeral (GH) joint is an incongruous, ball-and-socket (spheroidal) triaxial joint with a lax joint capsule. It is supported by the tendons of the rotator cuff and the glenohumeral (superior, middle, inferior) and coracohumeral ligaments (Fig. 17.2). The concave bony partner, the glenoid fossa, is located on the superior-lateral margin of the scapula. It faces anteriorly, laterally, and upward, which provides some stability to the joint. A fibrocartilagenous lip, the glenoid labrum, deepens the fossa for greater congruity and serves as the attachment site for the capsule. The convex bony partner is the head of the humerus. Only a small portion of the head comes in contact with the fossa at any one time, allowing for considerable humeral movement and potential instability.[133]

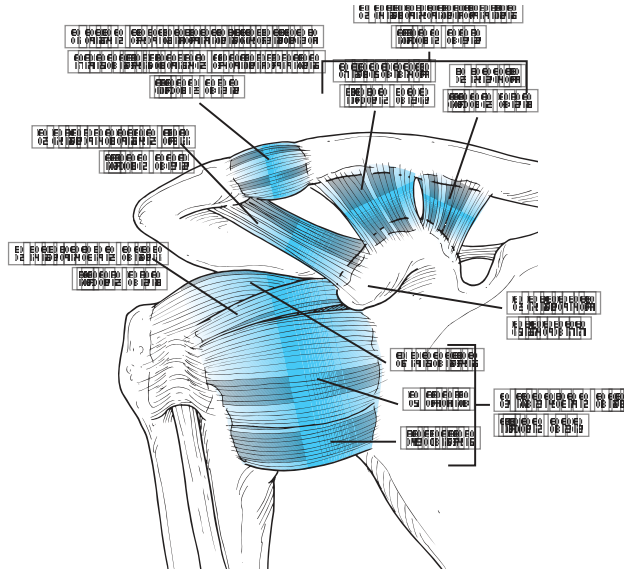


FIGURE 17.2 Ligaments of the glenohumeral and acromioclavicular joints.

Arthrokinematics

According to the convex-concave theory of joint motion (see Chapter 5), with motions of the humerus (physiologi-

BOX 17.1 Summary of Joint Arthrokinematics of the GH Joint

Physiological Motion of the Humerus	Roll	Slide
Flexion	Anterior	Posterior
Horizontal adduction	Anterior	Posterior
Internal rotation	Anterior	Posterior
Extension	Posterior	Anterior
Horizontal abduction	Posterior	Anterior
External rotation	Posterior	Anterior
Abduction	Superior	Inferior

cal motions) the convex head rolls in the same direction and slides in the opposite direction in the glenoid fossa (Box 17.1).

Focus on Evidence

Of interest and apparent contradiction of this theory, one study reported that through the mid-range of the arc of passive motion there is minimal displacement of the humeral head. However, beyond mid-range the overall displacement of the head in normal joints is anterior with shoulder flexion and posterior with shoulder extension.[69] This cadaveric study demonstrated that the integrity of the capsular ligamentous system influenced the displacement and that both hyper- and hypomobility of the capsule changed the overall displacement of the humeral head with passive range of motion (ROM).

In another study, Howel et al.,[84] using radiographs, measured humeral head displacement in normal and unstable shoulders. These investigators reported posterior displacement of the humeral head during end-range horizontal abduction with the humerus at 90° and in full external rotation in normal subjects, yet anterior displacement in subjects with anterior instability. These studies support the importance of joint mobility testing to examine restricted accessory motions to determine if interventions with joint mobilization techniques should be used and the direction of the mobilization force, rather than just using the convex-concave rule, to determine the direction of mobilizations.

Stability

Static and dynamic restraints provide joint stability (Table 17.1).[31,44,162,199,202] The structural relationship of the bony anatomy, ligaments, and glenoid labrum and the adhesive and cohesive forces in the joint provide static stability. The tendons of the rotator cuff blend with the ligaments and glenoid labrum at their sites of attachment so when the muscles contract they provide dynamic stability by tightening the static restraints (Fig. 17.3). The coordinated response of the muscles of the cuff and tension in the ligaments provide varying degrees of support depending on the position and motion of the humerus.[151,162,184] In addition, the long head of the biceps and the long head of the

TABLE 17.1 Static and Dynamic Stabilizers of the Scapula and Glenohumeral Joint

Description	Static Stabilizers	Dynamic Stabilizers
Scapula		
Weight of upper extremity creates downward rotation and forward tipping moment on the scapula	• Cohesive forces of subscapular bursa	Upper trapezius and serratus anterior middle trapezius and rhomboids
Glenohumeral joint		
In dependent position: if scapula is in normal alignment, weight of arm creates an adduction moment on the humerus	• Superior capsule and suprahumeral ligament are taut • Adhesive and cohesive forces of synovial fluid and negative joint pressure hold surfaces together • Glenoid labrum deepens fossa and improves congruency	Rotator cuff, deltoid, and long head of biceps brachii
When the humerus is elevating and the scapula is rotating upward	• Tension placed on static restraints by the rotator cuff • Glenohumeral ligaments provide inferior translation of humeral head	Rotator cuff and deltoid; elbow action brings in two-joint muscle support • Long head of biceps stabilizes against humeral elevation • Long head of triceps stabilizes against inferior translation

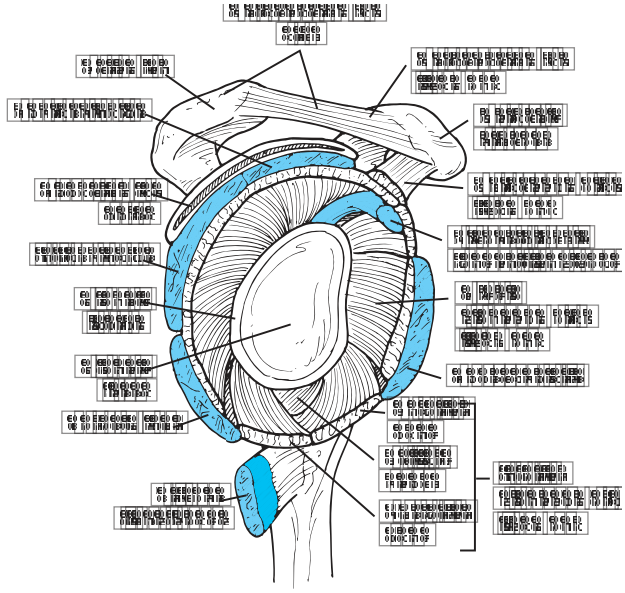


FIGURE 17.3 Lateral aspect of the glenoid fossa (interior view), showing attachments of the glenoid labrum, capsule, ligaments, and their relationship to the rotator cuff and long head of the biceps brachii musculature.

triceps brachii reinforce the capsule with their attachments and provide superior and inferior shoulder joint support, respectively, when functioning with elbow motions.[96] The long head of the biceps in particular stabilizes against humeral elevation[96] and contributes to anterior stability of the glenohumeral joint by resisting torsional forces when the shoulder is abducted and externally rotated.[11,151] Neuromuscular control, including movement awareness and motor response, underlies coordination of the dynamic restraints.[199,202]

Acromioclavicular Joint

The acromioclavicular (AC) joint is a plane, triaxial joint that may or may not have a disk. The weak capsule is reinforced by the superior and inferior AC ligaments (see Fig. 17.2). The convex bony partner is a facet on the lateral end of the clavicle. The concave bony partner is a facet on the acromion of the scapula.

Arthrokinematics

With motions of the scapula, the acromial surface slides in the same direction in which the scapula moves because the surface is concave. Motions affecting this joint include upward rotation (the scapula turns so the glenoid fossa rotates upward), downward rotation, winging of the vertebral border, and tipping of the inferior angle.

Stability

The AC ligaments are supported by the strong coracoclavicular ligament. No muscles directly cross this joint for dynamic support.

Sternoclavicular Joint

The sternoclavicular (SC) joint is an incongruent, triaxial, saddle-shaped joint with a disk. The joint is supported by

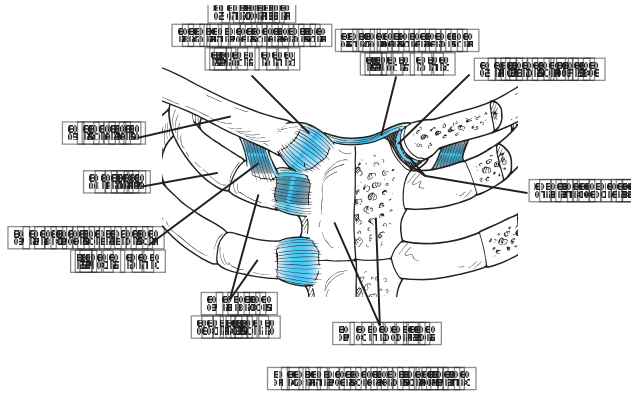


FIGURE 17.4 Ligaments of the sternoclavicular joint.

the anterior and posterior SC ligaments and the interclavicular and costoclavicular ligaments (Fig. 17.4). The medial end of the clavicle is convex superior to inferior and concave anterior to posterior. The joint disk attaches to the upper end. The superior-lateral portion of the manubrium and first costal cartilage is concave superior to inferior and convex anterior to posterior.

Arthrokinematics

The motions of the clavicle occur as a result of the scapular motions of elevation, depression, protraction (abduction), and retraction (adduction) (Box 17.2). Rotation of the clavicle occurs as an accessory motion when the humerus is elevated above the horizontal position and the scapula upwardly rotates; it cannot occur as an isolated voluntary motion.

Stability

The ligaments crossing the joint provide static stability. There are no muscles crossing the joint for dynamic stability.[41]

Functional Articulations

Scapulothoracic Articulation

Normally there is considerable soft tissue flexibility, allowing the scapula to slide along the thorax and participate in all upper extremity motions.

Motions of the Scapula

Motions of the scapula are:

BOX 17.2 Summary of Arthrokinematics of the SC Joint

Physiological Motion of the Clavicle	Roll	Slide
Protraction	Anterior	Anterior
Retraction	Posterior	Posterior
Elevation	Superior	Inferior
Depression	Inferior	Superior

- Elevation, depression, protraction (abduction), and retraction (adduction), seen with clavicular motions at the SC joint (Fig. 17.5 A&B). They are also component motions when the humerus moves.
- Upward and downward rotation, seen with clavicular motions at the SC joint and rotation at the AC joint, occurs concurrently with motions of the humerus (Fig. 17.5C). Upward rotation of the scapula is a necessary component motion for full ROM of flexion and abduction of the humerus.
- Winging and tipping, seen with motion at the AC joint concurrently with motions of the humerus (Fig. 17.5D). Winging is a transverse plane motion where the medial border lifts away from the rib cage; it normally occurs with horizontal adduction of the humerus. Forward tipping of the scapula occurs in conjunction with internal rotation and extension of the humerus when reaching the hand behind the back.

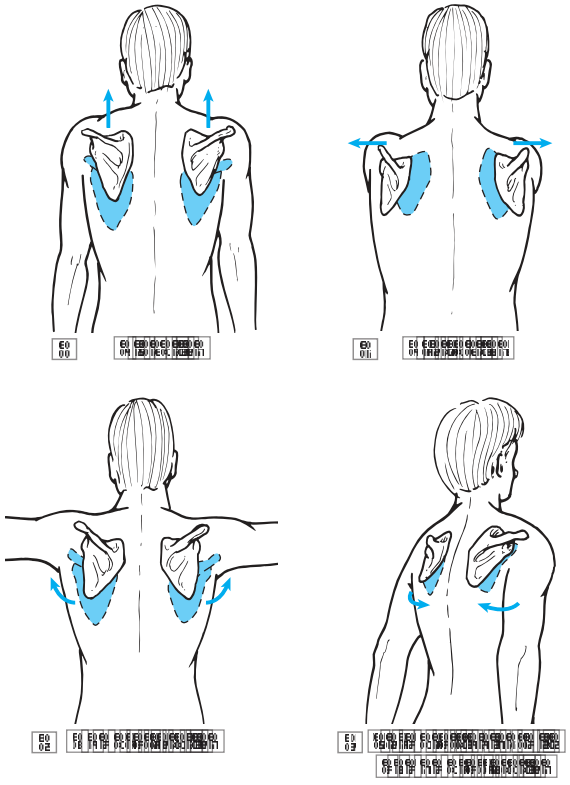


FIGURE 17.5 Scapular motions. (*A*) Elevation occurs with clavicular elevation at the SC joint when shrugging. (*B*) Protraction (abduction) occurs with clavicular abduction at the SC joint when reaching forward. (*C*) Upward rotation occurs with clavicular rotation at the SC and AC joints when flexing and abducting the shoulder. (*D*) Forward tipping (along with downward rotation) occurs at the AC joint when extending and internally rotating the shoulder.

Scapular Stability

Postural relationship. In the dependent position, the scapula is stabilized primarily through a balance of forces. The weight of the arm creates a downward rotation, abduction, and forward tipping moment on the scapula. The downward rotation is balanced by the dynamic support of the upper trapezius and serratus anterior. The forward tipping and abduction is balanced by the dynamic support of the rhomboids and middle trapezius[98,163] (see Table 17.1).

Active arm motions. With active arm motions, the muscles of the scapula function in synchrony to stabilize and control the position of the scapula so the scapulohumeral muscles can maintain an effective length–tension relationship as they function to stabilize and move the humerus. Without the positional control of the scapula, the efficiency of the humeral muscles decreases. The upper and lower trapezius with the serratus anterior upwardly rotate the scapula whenever the arm abducts or flexes, and the serratus anterior abducts (protracts) the scapula on the thorax to align the scapula during flexion or pushing activities. During arm extension or during pulling activities, the rhomboids function to downwardly rotate and adduct (retract) the scapula in synchrony with the latissimus dorsi, teres major, and rotator cuff muscles. These stabilizing muscles also eccentrically control acceleration motions of the scapula in the opposite directions.[135]

Faulty posture. With a faulty scapular posture, muscle length and strength imbalances occur not only in the scapular muscles but also in the humeral muscles, altering the mechanics of the glenohumeral joint. A forward tilt of the scapula (seen with a forward head posture and increased thoracic kyphosis) is associated with decreased flexibility in the pectoralis minor, levator scapulae, and scalenius muscles and weakness in the serratus anterior or trapezius muscles. This scapular posture changes the posture of the humerus in the glenoid, which assumes a relatively abducted and internally rotated position with respect to the scapula (Fig. 17.6). The glenohumeral internal rotators may become less flexible, and external rotators may weaken, affecting the mechanics of the joint.

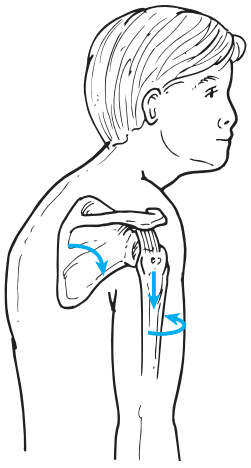


FIGURE 17.6 Faulty forward head, thoracic kyphosis, and shoulder girdle posture results in a forward tilt and downward rotation of the scapula with relative abduction and internal rotation of the humerus when the arm is in a dependent position.

Focus on Evidence

A study by Borstad and Ludewig,[19] which looked at the effect of pectoralis minor resting length on scapular kinematics in subjects without shoulder pain, documented that those individuals with a short pectoralis minor ($n = 25$) had greater scapular internal rotation (protraction) and less posterior tipping during arm elevation in flexion, abduction, and scaption than those with a longer pectoralis minor ($n = 25$), thus providing evidence for altered pectoralis minor muscle length and altered scapular movement. In a related study by the same author,[20] a correlation between the postural impairments of increased thoracic kyphosis, scapular internal rotation and forward tipping, and decreased pectoralis minor length was found to be significant, thus further supporting the relationship between muscle length and posture.

Suprahumeral (Subacromial) Space

The coracoacromial arch, composed of the acromion and coracoacromial ligament, overlies the subacromial/subdeltoid bursa, the supraspinatus tendon, and a portion of the muscle (Fig. 17.7).[98] These structures allow for and participate in normal shoulder function. Compromise of this space from faulty muscle function, faulty postural relationships, faulty joint mechanics, injury to the soft tissue in this region, or structural anomalies of the acromion lead to impingement syndromes.[19,27,30,99,103,113] After a rotator cuff tear, the bursa may communicate with the glenohumeral joint cavity.[44]

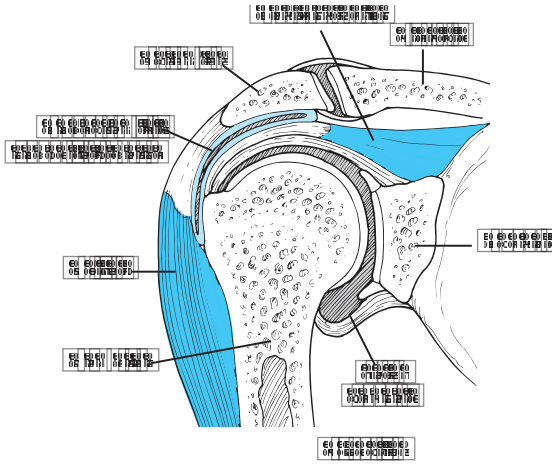


FIGURE 17.7 The supraspinatus and subacromial/subdeltoid bursa lie in the suprahumeral space.

Shoulder Girdle Function

Scapulohumeral Rhythm

Motion of the scapula, synchronous with motions of the humerus, allows for 150° to 180° of shoulder ROM into flexion or abduction with elevation. The ratio has considerable variation among individuals but is commonly accepted to be 2:1 (2° of glenohumeral motion to 1° of scapular rotation) overall motion. During the setting phase (0° to 30° abduction, 0° to 60° flexion), motion is primarily at the glenohumeral joint, whereas the scapula seeks a stable position. During the mid-range of humeral motion, the scapula has greater motion, approaching a 1:1 ratio with the humerus; later in the range, the glenohumeral joint again dominates the motion.[39,98,170]

- During humeral elevation, the synchronous motion of the scapula allows the muscles moving the humerus to maintain an effective length–tension relationship throughout the activity and helps maintain congruency between the humeral head and fossa while decreasing shear forces.[39,98,170]
- Muscles causing the upward rotation of the scapula are the upper and lower trapezius and the serratus anterior. Weakness or complete paralysis of these muscles results in the scapula being rotated downward by the contracting deltoid and supraspinatus as abduction or flexion is attempted. These two muscles then reach active insufficiency, and functional elevation of the arm cannot be reached, even though there may be normal passive ROM and normal strength in the shoulder abductor and flexor muscles.[170]
- During elevation of the humerus, the pectoralis minor is lengthened as the scapula upwardly rotates, retracts, and tips posteriorly. Restricted scapular movement during humeral elevation from a shortened pectoralis minor results in patterns similar to those seen in patients with impingement symptoms and could be a risk factor for development of the syndrome.[19]

Clavicular Elevation and Rotation With Humeral Motion

It is commonly accepted that initially the first 30° of upward rotation of the scapula occurs with elevation of the clavicle at the SC joint. Then, as the coracoclavicular ligament becomes taut, the clavicle rotates 38° to 55° about its longitudinal axis, which elevates its acromial end (because it is crank-shaped). This motion allows the scapula to rotate an additional 30° at the AC joint.[98] Loss of any of these functional components decreases the amount of scapular rotation and thus the ROM of the upper extremity.

Focus on Evidence

A recent three-dimensional study of clavicular motion during humeral flexion, scaption, and abduction to 115° using surface electromagnetic sensors on 30 asymptomatic subjects and 9 individuals with shoulder pathology documented 11° to 15° of clavicular elevation, 15° to 29° of retraction, and 15° to 31° of posterior long axis rotation, showing similar patterns but different ranges from previously reported studies.[104] Ranges of clavicular motion